Captstone Data 205

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2026-04-20

#Librarires Laoding

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(tidycensus)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

#ZHVI County Csv

```
housing <- read.csv(
  "~/Downloads/County_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv",
  check.names = FALSE
)
```

#Zillow City data

```r
housing_city <- read.csv(
  "~/Downloads/City_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv",
  check.names = FALSE
)
```

#Cencus API key

```r
census_api_key("1f91d77814ba22215a92913930a3331723837cb3",overwrite=TRUE)
```

```
## To install your API key for use in future sessions, run this function with `install = TRUE`.
```

#Filter Montgomery County cities

```r
#Creating variable to store all of MoCO cities into
moco_cities <- housing_city %>%
  filter(StateName == "MD", CountyName == "Montgomery County")

#outputting all cities in MoCo
unique(moco_cities$RegionName)
```

```
##  [1] "Silver Spring"      "Rockville"          "Gaithersburg"
##  [4] "Germantown"         "Bethesda"           "Potomac"
##  [7] "Montgomery Village" "Takoma Park"        "Chevy Chase"
## [10] "North Potomac"      "Olney"              "Derwood"
## [13] "Clarksburg"         "Burtonsville"       "Kensington"
## [16] "Damascus"           "Boyds"              "Brookeville"
## [19] "Darnestown"         "Poolesville"        "Sandy Spring"
## [22] "Dickerson"          "Cabin John"         "Ashton"
## [25] "Chevy Chase View"   "Garrett Park"       "Laytonsville"
## [28] "Washington Grove"   "Spencerville"       "Brinklow"
## [31] "Glen Echo"
```

#Data inspection

```r
glimpse(moco_cities)
```

```
## Rows: 31
## Columns: 323
## $ RegionID     <int> 7081, 33714, 45383, 31732, 37406, 47210, 19445, 47942, 39~
## $ SizeRank     <int> 79, 421, 425, 587, 652, 1289, 1883, 2438, 2519, 2520, 271~
## $ RegionName   <chr> "Silver Spring", "Rockville", "Gaithersburg", "Germantown~
## $ RegionType   <chr> "city", "city", "city", "city", "city", "city", "city", "~
## $ StateName    <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD~
## $ State        <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD~
## $ Metro        <chr> "Washington-Arlington-Alexandria, DC-VA-MD-WV", "Washingt~
## $ CountyName   <chr> "Montgomery County", "Montgomery County", "Montgomery Cou~
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## $ `2000-02-29` <dbl> 185642.1, 210734.8, 190413.3, 150640.0, 382635.3, 526249.~
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## $ `2000-04-30` <dbl> 185978.9, 211406.6, 190625.1, 150611.1, 384784.5, 529148.~
## $ `2000-05-31` <dbl> 186870.6, 212623.3, 191449.0, 151133.7, 387763.5, 533296.~
```

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## $ `2026-03-31` <dbl> 550712.4, 618892.3, 527608.4, 423001.5, 1152218.6, 139021~
```

```
colnames(moco_cities)
```

```
##   [1] "RegionID"   "SizeRank"   "RegionName" "RegionType" "StateName"
##   [6] "State"      "Metro"      "CountyName" "2000-01-31" "2000-02-29"
##  [11] "2000-03-31" "2000-04-30" "2000-05-31" "2000-06-30" "2000-07-31"
##  [16] "2000-08-31" "2000-09-30" "2000-10-31" "2000-11-30" "2000-12-31"
##  [21] "2001-01-31" "2001-02-28" "2001-03-31" "2001-04-30" "2001-05-31"
##  [26] "2001-06-30" "2001-07-31" "2001-08-31" "2001-09-30" "2001-10-31"
##  [31] "2001-11-30" "2001-12-31" "2002-01-31" "2002-02-28" "2002-03-31"
##  [36] "2002-04-30" "2002-05-31" "2002-06-30" "2002-07-31" "2002-08-31"
##  [41] "2002-09-30" "2002-10-31" "2002-11-30" "2002-12-31" "2003-01-31"
##  [46] "2003-02-28" "2003-03-31" "2003-04-30" "2003-05-31" "2003-06-30"
##  [51] "2003-07-31" "2003-08-31" "2003-09-30" "2003-10-31" "2003-11-30"
```

```
##  [56] "2003-12-31" "2004-01-31" "2004-02-29" "2004-03-31" "2004-04-30"
##  [61] "2004-05-31" "2004-06-30" "2004-07-31" "2004-08-31" "2004-09-30"
##  [66] "2004-10-31" "2004-11-30" "2004-12-31" "2005-01-31" "2005-02-28"
##  [71] "2005-03-31" "2005-04-30" "2005-05-31" "2005-06-30" "2005-07-31"
##  [76] "2005-08-31" "2005-09-30" "2005-10-31" "2005-11-30" "2005-12-31"
##  [81] "2006-01-31" "2006-02-28" "2006-03-31" "2006-04-30" "2006-05-31"
##  [86] "2006-06-30" "2006-07-31" "2006-08-31" "2006-09-30" "2006-10-31"
##  [91] "2006-11-30" "2006-12-31" "2007-01-31" "2007-02-28" "2007-03-31"
##  [96] "2007-04-30" "2007-05-31" "2007-06-30" "2007-07-31" "2007-08-31"
## [101] "2007-09-30" "2007-10-31" "2007-11-30" "2007-12-31" "2008-01-31"
## [106] "2008-02-29" "2008-03-31" "2008-04-30" "2008-05-31" "2008-06-30"
## [111] "2008-07-31" "2008-08-31" "2008-09-30" "2008-10-31" "2008-11-30"
## [116] "2008-12-31" "2009-01-31" "2009-02-28" "2009-03-31" "2009-04-30"
## [121] "2009-05-31" "2009-06-30" "2009-07-31" "2009-08-31" "2009-09-30"
## [126] "2009-10-31" "2009-11-30" "2009-12-31" "2010-01-31" "2010-02-28"
## [131] "2010-03-31" "2010-04-30" "2010-05-31" "2010-06-30" "2010-07-31"
## [136] "2010-08-31" "2010-09-30" "2010-10-31" "2010-11-30" "2010-12-31"
## [141] "2011-01-31" "2011-02-28" "2011-03-31" "2011-04-30" "2011-05-31"
## [146] "2011-06-30" "2011-07-31" "2011-08-31" "2011-09-30" "2011-10-31"
## [151] "2011-11-30" "2011-12-31" "2012-01-31" "2012-02-29" "2012-03-31"
## [156] "2012-04-30" "2012-05-31" "2012-06-30" "2012-07-31" "2012-08-31"
## [161] "2012-09-30" "2012-10-31" "2012-11-30" "2012-12-31" "2013-01-31"
## [166] "2013-02-28" "2013-03-31" "2013-04-30" "2013-05-31" "2013-06-30"
## [171] "2013-07-31" "2013-08-31" "2013-09-30" "2013-10-31" "2013-11-30"
## [176] "2013-12-31" "2014-01-31" "2014-02-28" "2014-03-31" "2014-04-30"
## [181] "2014-05-31" "2014-06-30" "2014-07-31" "2014-08-31" "2014-09-30"
## [186] "2014-10-31" "2014-11-30" "2014-12-31" "2015-01-31" "2015-02-28"
## [191] "2015-03-31" "2015-04-30" "2015-05-31" "2015-06-30" "2015-07-31"
## [196] "2015-08-31" "2015-09-30" "2015-10-31" "2015-11-30" "2015-12-31"
## [201] "2016-01-31" "2016-02-29" "2016-03-31" "2016-04-30" "2016-05-31"
## [206] "2016-06-30" "2016-07-31" "2016-08-31" "2016-09-30" "2016-10-31"
## [211] "2016-11-30" "2016-12-31" "2017-01-31" "2017-02-28" "2017-03-31"
## [216] "2017-04-30" "2017-05-31" "2017-06-30" "2017-07-31" "2017-08-31"
## [221] "2017-09-30" "2017-10-31" "2017-11-30" "2017-12-31" "2018-01-31"
## [226] "2018-02-28" "2018-03-31" "2018-04-30" "2018-05-31" "2018-06-30"
## [231] "2018-07-31" "2018-08-31" "2018-09-30" "2018-10-31" "2018-11-30"
## [236] "2018-12-31" "2019-01-31" "2019-02-28" "2019-03-31" "2019-04-30"
## [241] "2019-05-31" "2019-06-30" "2019-07-31" "2019-08-31" "2019-09-30"
## [246] "2019-10-31" "2019-11-30" "2019-12-31" "2020-01-31" "2020-02-29"
## [251] "2020-03-31" "2020-04-30" "2020-05-31" "2020-06-30" "2020-07-31"
## [256] "2020-08-31" "2020-09-30" "2020-10-31" "2020-11-30" "2020-12-31"
## [261] "2021-01-31" "2021-02-28" "2021-03-31" "2021-04-30" "2021-05-31"
## [266] "2021-06-30" "2021-07-31" "2021-08-31" "2021-09-30" "2021-10-31"
## [271] "2021-11-30" "2021-12-31" "2022-01-31" "2022-02-28" "2022-03-31"
## [276] "2022-04-30" "2022-05-31" "2022-06-30" "2022-07-31" "2022-08-31"
## [281] "2022-09-30" "2022-10-31" "2022-11-30" "2022-12-31" "2023-01-31"
## [286] "2023-02-28" "2023-03-31" "2023-04-30" "2023-05-31" "2023-06-30"
## [291] "2023-07-31" "2023-08-31" "2023-09-30" "2023-10-31" "2023-11-30"
## [296] "2023-12-31" "2024-01-31" "2024-02-29" "2024-03-31" "2024-04-30"
## [301] "2024-05-31" "2024-06-30" "2024-07-31" "2024-08-31" "2024-09-30"
## [306] "2024-10-31" "2024-11-30" "2024-12-31" "2025-01-31" "2025-02-28"
## [311] "2025-03-31" "2025-04-30" "2025-05-31" "2025-06-30" "2025-07-31"
## [316] "2025-08-31" "2025-09-30" "2025-10-31" "2025-11-30" "2025-12-31"
## [321] "2026-01-31" "2026-02-28" "2026-03-31"
```

#Clean Cencus Data

```
census_data <- get_acs(
  geography = "place",
  state = "MD",
  variables = c(
    population = "B01003_001",
    median_income = "B19013_001",
    census_home_value = "B25077_001",
    median_rent = "B25064_001"
  ),
  year = 2022,
  survey = "acs5"
)
```

```
## Getting data from the 2018-2022 5-year ACS
```

```
census_clean <- census_data %>%
  select(NAME, variable, estimate) %>%
  pivot_wider(names_from = variable, values_from = estimate) %>%
  mutate(
    RegionName = NAME,
    RegionName = str_remove(RegionName, " city, Maryland"),
    RegionName = str_remove(RegionName, " town, Maryland"),
    RegionName = str_remove(RegionName, " village, Maryland"),
    RegionName = str_remove(RegionName, " CDP, Maryland")
  ) %>%
  select(RegionName, population, median_income, census_home_value, median_rent)
```

#Reshape data

```
#Reshape Zillow city data from wide to long format
moco_long <- moco_cities %>%
  pivot_longer(
    cols = matches("^\\d{4}-\\d{2}-\\d{2}$|^X\\d{4}\\.\\d{2}\\.\\d{2}$"),
    names_to = "Date",
    values_to = "ZHVI"
  ) %>%
  mutate(
    Date = str_remove(Date, "^X"),
    Date = str_replace_all(Date, "\\.", "-"),
    Date = as.Date(Date)
  )
```

#Post cleaning inspection

```
glimpse(moco_long)
```

```
## Rows: 9,765
## Columns: 10
## $ RegionID   <int> 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081, 7081,~
## $ SizeRank   <int> 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79, 79,~
```

```
## $ RegionName <chr> "Silver Spring", "Silver Spring", "Silver Spring", "Silver ~
## $ RegionType <chr> "city", "city", "city", "city", "city", "city", "city", "ci~
## $ StateName  <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD",~
## $ State      <chr> "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD", "MD",~
## $ Metro      <chr> "Washington-Arlington-Alexandria, DC-VA-MD-WV", "Washington~
## $ CountyName <chr> "Montgomery County", "Montgomery County", "Montgomery Count~
## $ Date       <date> 2000-01-31, 2000-02-29, 2000-03-31, 2000-04-30, 2000-05-31~
## $ ZHVI       <dbl> 185818.2, 185642.1, 185688.0, 185978.9, 186870.6, 188282.9,~
```

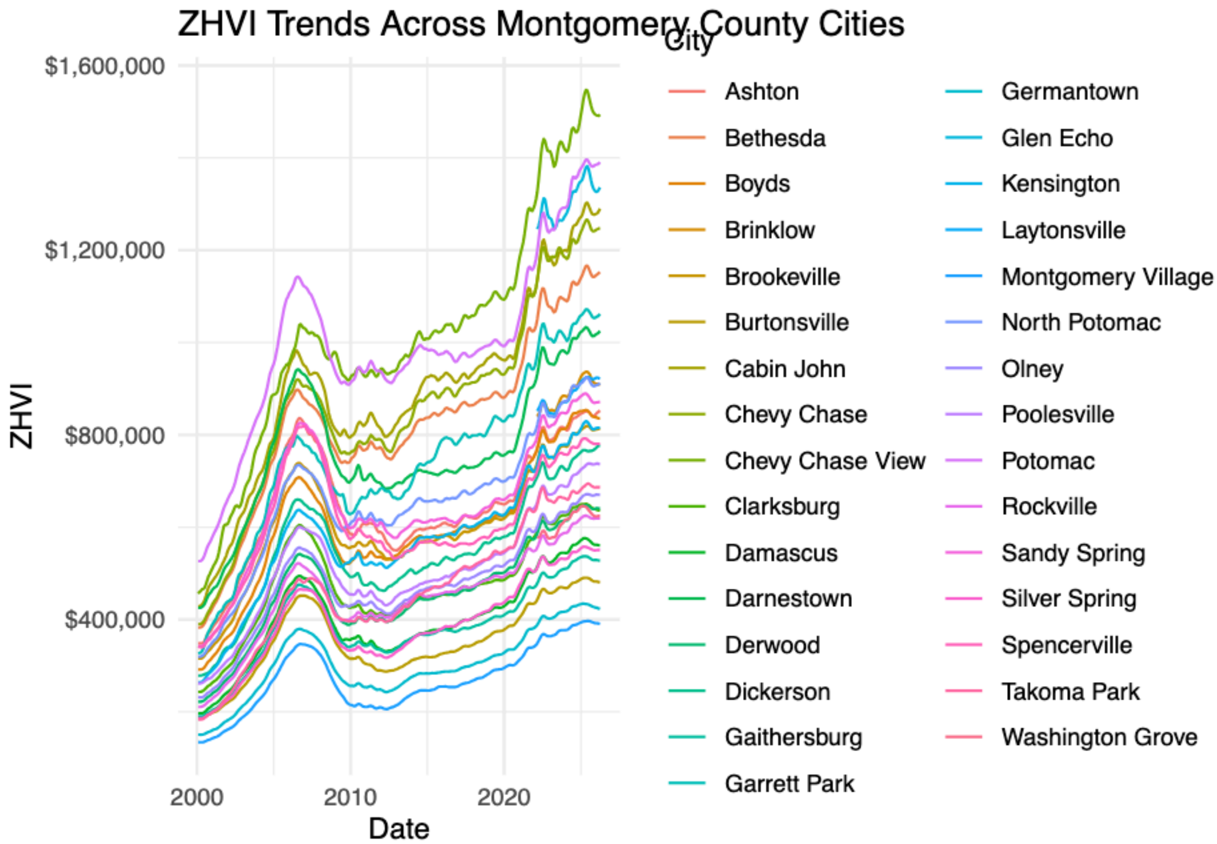
```r
summary(moco_long$ZHVI)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  134181  414426  564403  599572  756806 1547109    1060
```

#Time series visualization

```r
#Home value trends across Montgomery County cities
ggplot(moco_long, aes(x = Date, y = ZHVI, color = RegionName)) +
  geom_line() +
  scale_y_continuous(labels = dollar_format()) +
  labs(
    title = "ZHVI Trends Across Montgomery County Cities",
    x = "Date",
    y = "ZHVI",
    color = "City"
  ) +
  theme_minimal()
```

```
## Warning: Removed 1060 rows containing missing values or values outside the scale range
## (`geom_line()`).
```



#Latest prices calculation

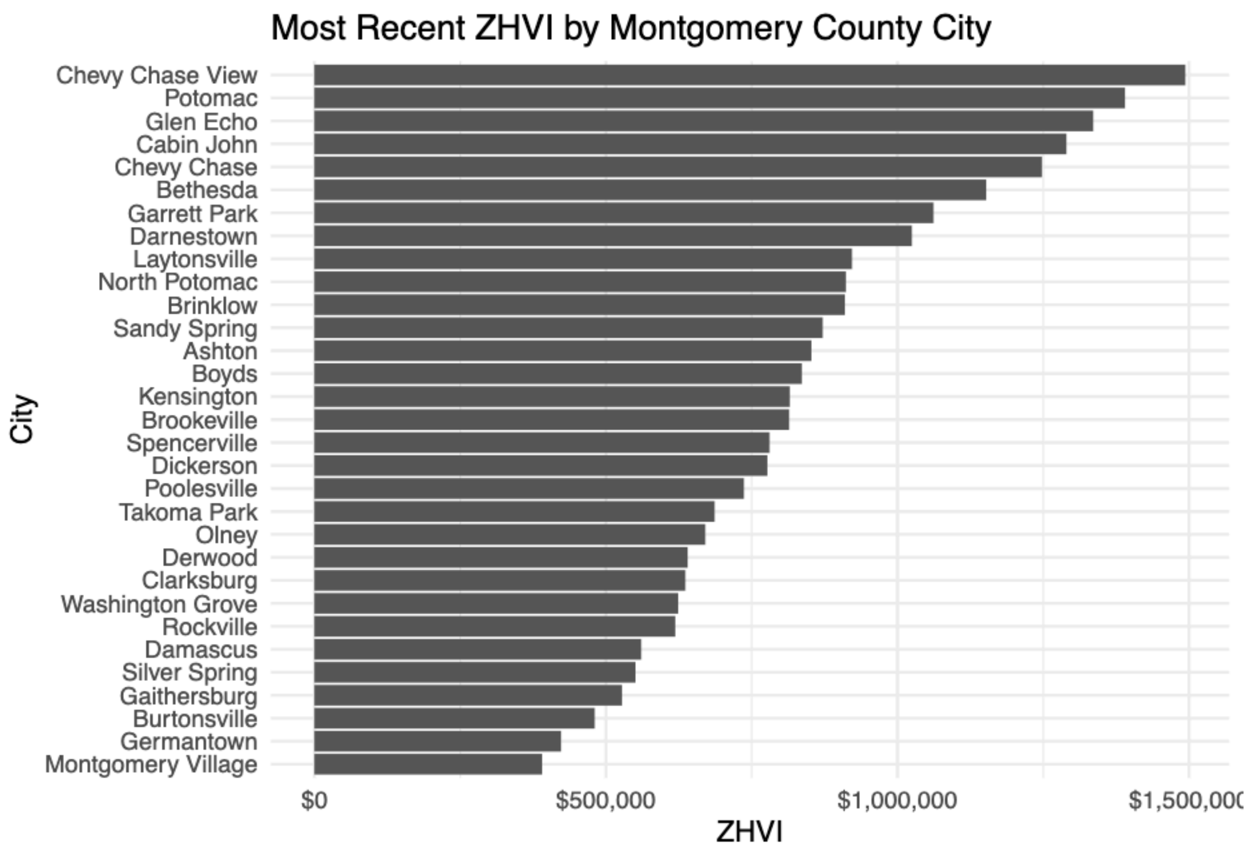
```r
#Creating new Variable of latest pricing
latest_prices <- moco_long %>%
  group_by(RegionName) %>%
  filter(Date == max(Date, na.rm = TRUE)) %>%
  ungroup() %>%
  select(RegionName, Date, ZHVI) %>%
  arrange(desc(ZHVI))

latest_prices
```

```
## # A tibble: 31 x 3
##    RegionName       Date          ZHVI
##    <chr>            <date>       <dbl>
##  1 Chevy Chase View 2026-03-31 1493700.
##  2 Potomac          2026-03-31 1390212.
##  3 Glen Echo        2026-03-31 1335696.
##  4 Cabin John       2026-03-31 1289768.
##  5 Chevy Chase      2026-03-31 1247761.
##  6 Bethesda         2026-03-31 1152219.
##  7 Garrett Park     2026-03-31 1061769.
##  8 Darnestown       2026-03-31 1024744.
##  9 Laytonsville     2026-03-31  922014.
## 10 North Potomac    2026-03-31  911608.
## # i 21 more rows
```

#Bar chart of latest prices

```r
ggplot(latest_prices, aes(x = reorder(RegionName, ZHVI), y = ZHVI)) +
  geom_col() +
  coord_flip() +
  scale_y_continuous(labels = dollar_format()) +
  labs(
    title = "Most Recent ZHVI by Montgomery County City",
    x = "City",
    y = "ZHVI"
  ) +
  theme_minimal()
```



Calculating the long term growth by the city

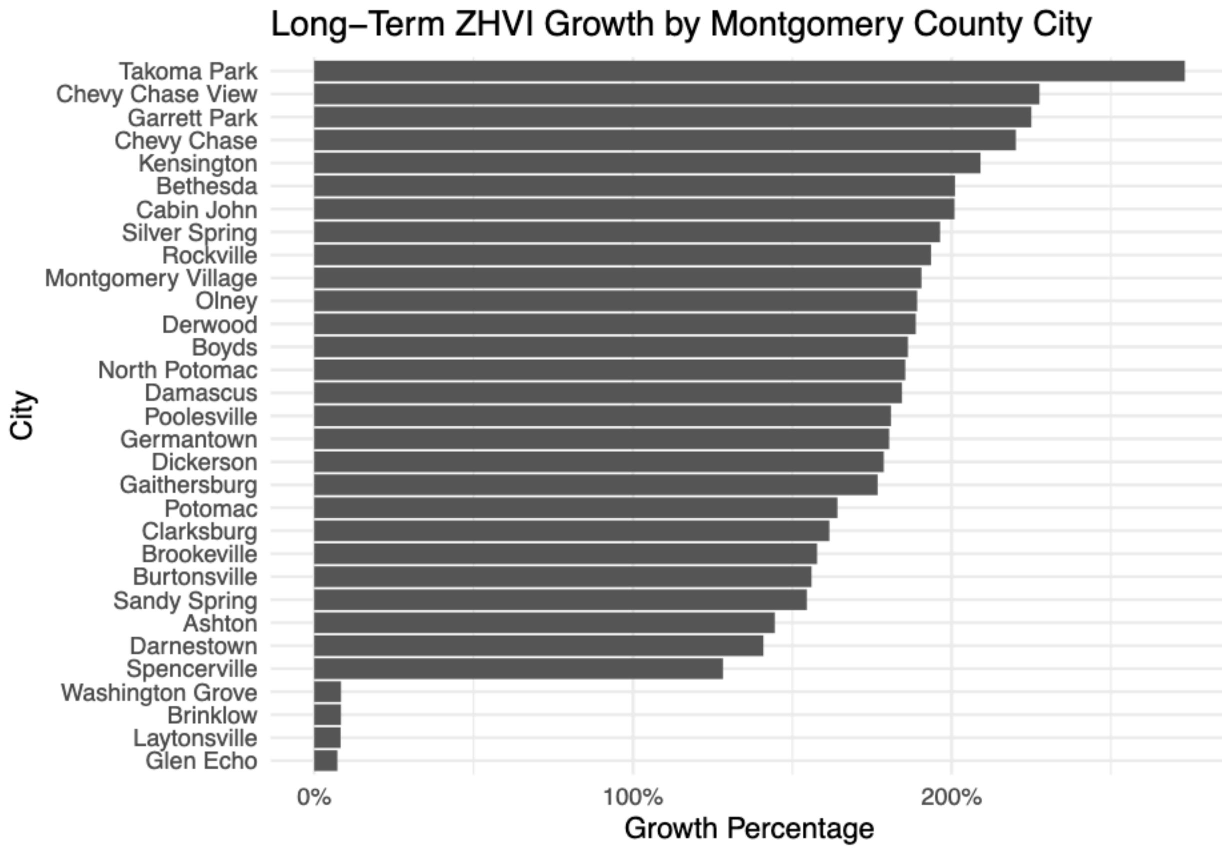
```r
# Calculate long-term growth by city
growth_summary <- moco_long %>%
  filter(!is.na(ZHVI)) %>%
  group_by(RegionName) %>%
  summarise(
    start_date = min(Date),
    end_date = max(Date),
    start_price = ZHVI[which.min(Date)],
    end_price = ZHVI[which.max(Date)],
    growth_dollars = end_price - start_price,
    growth_percent = (end_price - start_price) / start_price
  ) %>%
```

```r
  arrange(desc(growth_percent))

growth_summary
```

```
## # A tibble: 31 x 7
##    RegionName          start_date end_date   start_price end_price growth_dollars
##    <chr>               <date>     <date>           <dbl>     <dbl>          <dbl>
##  1 Takoma Park         2000-01-31 2026-03-31     183931.   686399.        502468.
##  2 Chevy Chase View    2000-01-31 2026-03-31     456026.  1493700.       1037674.
##  3 Garrett Park        2000-01-31 2026-03-31     326681.  1061769.        735088.
##  4 Chevy Chase         2000-01-31 2026-03-31     389743.  1247761.        858019.
##  5 Kensington          2000-01-31 2026-03-31     263815.   815355.        551540.
##  6 Bethesda            2000-01-31 2026-03-31     382718.  1152219.        769501.
##  7 Cabin John          2000-01-31 2026-03-31     428641.  1289768.        861127.
##  8 Silver Spring       2000-01-31 2026-03-31     185818.   550712.        364894.
##  9 Rockville           2000-01-31 2026-03-31     210841.   618892.        408051.
## 10 Montgomery Village  2000-01-31 2026-03-31     134430.   390564.        256134.
## # i 21 more rows
## # i 1 more variable: growth_percent <dbl>
```

```r
ggplot(growth_summary, aes(x = reorder(RegionName, growth_percent), y = growth_percent)) +
  geom_col() +
  coord_flip() +
  scale_y_continuous(labels = percent_format()) +
  labs(
    title = "Long-Term ZHVI Growth by Montgomery County City",
    x = "City",
    y = "Growth Percentage"
  ) +
  theme_minimal()
```




```r
# Calculate volatility using monthly percent changes
volatility_summary <- moco_long %>%
  filter(!is.na(ZHVI)) %>%
  arrange(RegionName, Date) %>%
  group_by(RegionName) %>%
  mutate(monthly_change = (ZHVI - lag(ZHVI)) / lag(ZHVI)) %>%
  summarise(
    avg_monthly_change = mean(monthly_change, na.rm = TRUE),
    volatility = sd(monthly_change, na.rm = TRUE)
  ) %>%
  arrange(desc(volatility))

volatility_summary
```

```
## # A tibble: 31 x 3
##    RegionName         avg_monthly_change volatility
##    <chr>                           <dbl>      <dbl>
##  1 Montgomery Village            0.00344    0.00868
##  2 Garrett Park                  0.00379    0.00795
##  3 Germantown                    0.00332    0.00793
##  4 Ashton                        0.00288    0.00786
##  5 Spencerville                  0.00266    0.00785
##  6 Damascus                      0.00336    0.00780
##  7 Dickerson                     0.00330    0.00770
##  8 Silver Spring                 0.00350    0.00767
##  9 Clarksburg                    0.00310    0.00758
```

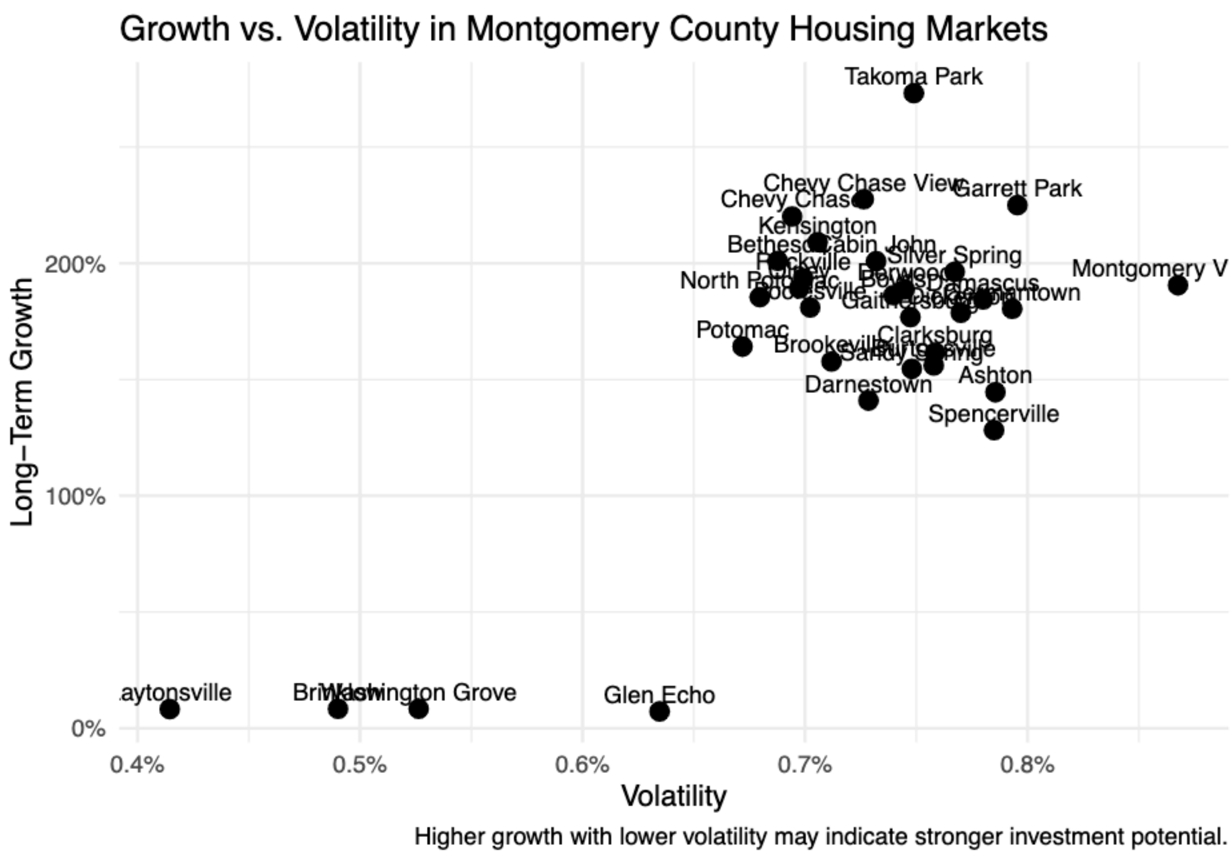
```
## 10 Burtonsville                    0.00303    0.00758
## # i 21 more rows
```

```r
investment_summary <- growth_summary %>%
  left_join(volatility_summary, by = "RegionName") %>%
  arrange(desc(growth_percent))

investment_summary
```

```
## # A tibble: 31 x 9
##    RegionName         start_date end_date   start_price end_price growth_dollars
##    <chr>              <date>     <date>           <dbl>     <dbl>          <dbl>
##  1 Takoma Park        2000-01-31 2026-03-31     183931.   686399.        502468.
##  2 Chevy Chase View   2000-01-31 2026-03-31     456026.  1493700.       1037674.
##  3 Garrett Park       2000-01-31 2026-03-31     326681.  1061769.        735088.
##  4 Chevy Chase        2000-01-31 2026-03-31     389743.  1247761.        858019.
##  5 Kensington         2000-01-31 2026-03-31     263815.   815355.        551540.
##  6 Bethesda           2000-01-31 2026-03-31     382718.  1152219.        769501.
##  7 Cabin John         2000-01-31 2026-03-31     428641.  1289768.        861127.
##  8 Silver Spring      2000-01-31 2026-03-31     185818.   550712.        364894.
##  9 Rockville          2000-01-31 2026-03-31     210841.   618892.        408051.
## 10 Montgomery Village 2000-01-31 2026-03-31     134430.   390564.        256134.
## # i 21 more rows
## # i 3 more variables: growth_percent <dbl>, avg_monthly_change <dbl>,
## #   volatility <dbl>
```

```r
ggplot(investment_summary, aes(x = volatility, y = growth_percent, label = RegionName)) +
  geom_point(size = 3) +
  geom_text(vjust = -0.5, size = 3) +
  scale_y_continuous(labels = percent_format()) +
  scale_x_continuous(labels = percent_format()) +
  labs(
    title = "Growth vs. Volatility in Montgomery County Housing Markets",
    x = "Volatility",
    y = "Long-Term Growth",
    caption = "Higher growth with lower volatility may indicate stronger investment potential."
  ) +
  theme_minimal()
```




```r
investment_summary <- investment_summary %>%
  mutate(
    risk_adjusted_return = growth_percent / volatility
  ) %>%
  arrange(desc(risk_adjusted_return))

investment_summary
```

```
## # A tibble: 31 x 10
##    RegionName        start_date end_date   start_price end_price growth_dollars
##    <chr>             <date>     <date>           <dbl>     <dbl>          <dbl>
##  1 Takoma Park       2000-01-31 2026-03-31     183931.   686399.        502468.
##  2 Chevy Chase       2000-01-31 2026-03-31     389743.  1247761.        858019.
##  3 Chevy Chase View  2000-01-31 2026-03-31     456026.  1493700.       1037674.
##  4 Kensington        2000-01-31 2026-03-31     263815.   815355.        551540.
##  5 Bethesda          2000-01-31 2026-03-31     382718.  1152219.        769501.
##  6 Garrett Park      2000-01-31 2026-03-31     326681.  1061769.        735088.
##  7 Rockville         2000-01-31 2026-03-31     210841.   618892.        408051.
##  8 Cabin John        2000-01-31 2026-03-31     428641.  1289768.        861127.
##  9 North Potomac     2000-01-31 2026-03-31     319353.   911608.        592255.
## 10 Olney             2000-01-31 2026-03-31     231814.   670386.        438572.
## # i 21 more rows
## # i 4 more variables: growth_percent <dbl>, avg_monthly_change <dbl>,
## #   volatility <dbl>, risk_adjusted_return <dbl>
```

```r
combined_data <- latest_prices %>%
  left_join(census_clean, by = "RegionName")
```

```r
growth_with_demo <- growth_summary %>%
  left_join(census_clean, by = "RegionName")
```

```r
growth_with_demo %>%
  select(RegionName, growth_percent, population, median_income) %>%
  print(n = 50)
```

```
## # A tibble: 32 x 4
##    RegionName          growth_percent population median_income
##    <chr>                        <dbl>      <dbl>         <dbl>
##  1 Takoma Park                 2.73        17542         95316
##  2 Chevy Chase View            2.28         1123        250001
##  3 Garrett Park                2.25          816        250001
##  4 Chevy Chase                 2.20         2879        250001
##  5 Chevy Chase                 2.20         9789        217500
##  6 Kensington                  2.09         2241        143021
##  7 Bethesda                    2.01        66316        185546
##  8 Cabin John                  2.01         2196        208598
##  9 Silver Spring               1.96        81808         95213
## 10 Rockville                   1.94        67142        122470
## 11 Montgomery Village          1.91        34748         91703
## 12 Olney                       1.89        35522        166541
## 13 Derwood                     1.89         1850         77083
## 14 Boyds                       1.86           NA            NA
## 15 North Potomac               1.85        24692        182083
## 16 Damascus                    1.84        16846        149234
## 17 Poolesville                 1.81         5686        215025
## 18 Germantown                  1.80        90210        109268
## 19 Dickerson                   1.79           NA            NA
## 20 Gaithersburg                1.77        69016        104544
## 21 Potomac                     1.64        46499        218710
## 22 Clarksburg                  1.62        28337        165551
## 23 Brookeville                 1.58          164        182188
## 24 Burtonsville                1.56        10316        138416
## 25 Sandy Spring                1.55           NA            NA
## 26 Ashton                      1.45           NA            NA
## 27 Darnestown                  1.41         6672        250001
## 28 Spencerville                1.28         1277        107019
## 29 Washington Grove            0.0838        604        131528
## 30 Brinklow                    0.0837         NA            NA
## 31 Laytonsville                0.0827        736        200625
## 32 Glen Echo                   0.0725        282        194375
```

```r
census_clean <- census_data %>%
  select(NAME, variable, estimate) %>%
  pivot_wider(names_from = variable, values_from = estimate) %>%
  mutate(
    RegionName = NAME,
    RegionName = str_remove(RegionName, " city, Maryland"),
```

```r
    RegionName = str_remove(RegionName, " town, Maryland"),
    RegionName = str_remove(RegionName, " village, Maryland"),
    RegionName = str_remove(RegionName, " CDP, Maryland")
  ) %>%
  select(RegionName, population, median_income)
```

```r
growth_with_demo <- growth_summary %>%
  left_join(census_clean, by = "RegionName")
```

```r
growth_with_demo %>%
  select(RegionName, growth_percent, population, median_income) %>%
  print(n = 50)
```

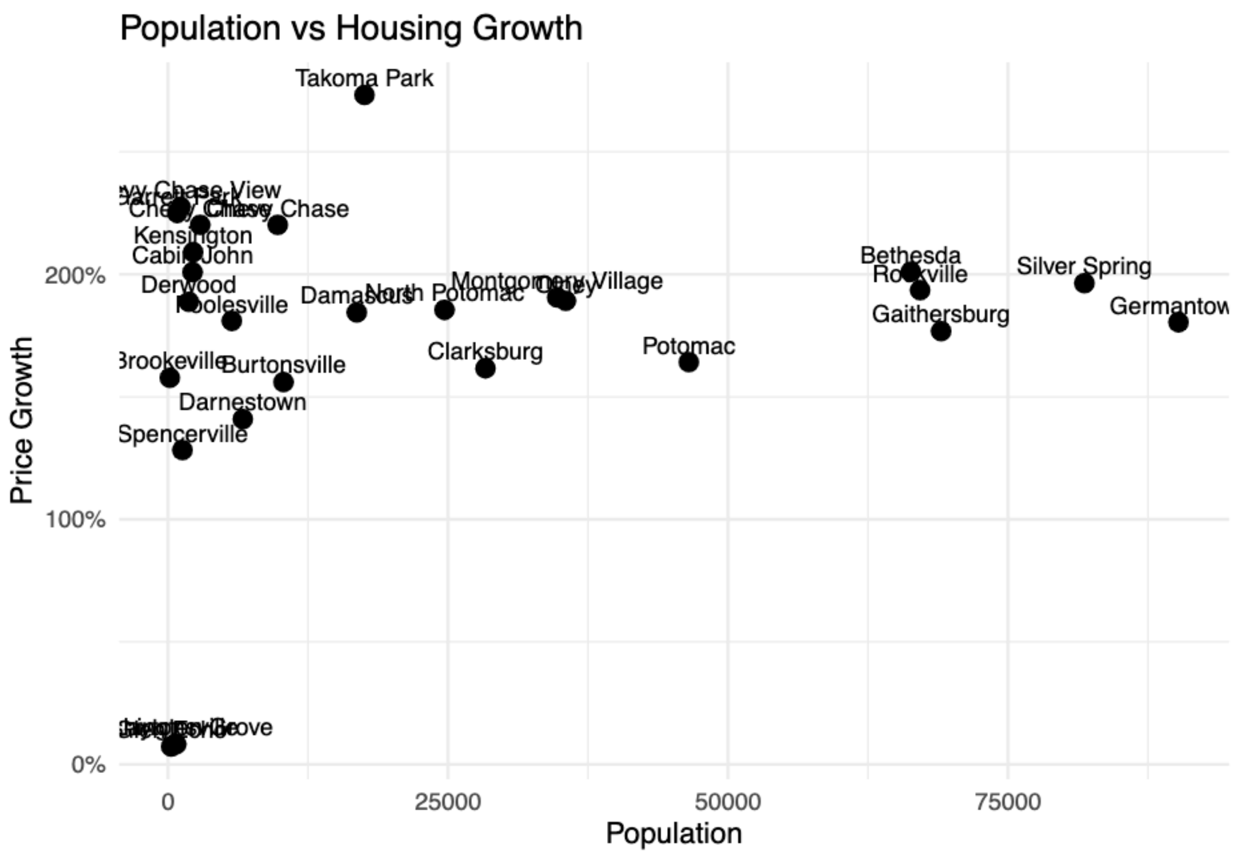
```
## # A tibble: 32 x 4
##    RegionName          growth_percent population median_income
##    <chr>                        <dbl>      <dbl>         <dbl>
##  1 Takoma Park                 2.73        17542         95316
##  2 Chevy Chase View            2.28         1123        250001
##  3 Garrett Park                2.25          816        250001
##  4 Chevy Chase                 2.20         2879        250001
##  5 Chevy Chase                 2.20         9789        217500
##  6 Kensington                  2.09         2241        143021
##  7 Bethesda                    2.01        66316        185546
##  8 Cabin John                  2.01         2196        208598
##  9 Silver Spring               1.96        81808         95213
## 10 Rockville                   1.94        67142        122470
## 11 Montgomery Village          1.91        34748         91703
## 12 Olney                       1.89        35522        166541
## 13 Derwood                     1.89         1850         77083
## 14 Boyds                       1.86           NA            NA
## 15 North Potomac               1.85        24692        182083
## 16 Damascus                    1.84        16846        149234
## 17 Poolesville                 1.81         5686        215025
## 18 Germantown                  1.80        90210        109268
## 19 Dickerson                   1.79           NA            NA
## 20 Gaithersburg                1.77        69016        104544
## 21 Potomac                     1.64        46499        218710
## 22 Clarksburg                  1.62        28337        165551
## 23 Brookeville                 1.58          164        182188
## 24 Burtonsville                1.56        10316        138416
## 25 Sandy Spring                1.55           NA            NA
## 26 Ashton                      1.45           NA            NA
## 27 Darnestown                  1.41         6672        250001
## 28 Spencerville                1.28         1277        107019
## 29 Washington Grove            0.0838        604        131528
## 30 Brinklow                    0.0837         NA            NA
## 31 Laytonsville                0.0827        736        200625
## 32 Glen Echo                   0.0725        282        194375
```

```r
ggplot(growth_with_demo, aes(x = population, y = growth_percent, label = RegionName)) +
  geom_point(size = 3) +
  geom_text(vjust = -0.5, size = 3) +
```

```r
  scale_y_continuous(labels = percent_format()) +
  labs(
    title = "Population vs Housing Growth",
    x = "Population",
    y = "Price Growth"
  ) +
  theme_minimal()
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_text()`).
```




```r
# Pull Census ACS data for Maryland places
# This gives population, income, home value, and rent estimates
census_data <- get_acs(
  geography = "place",
  state = "MD",
  variables = c(
    population = "B01003_001",
    median_income = "B19013_001",
    census_home_value = "B25077_001",
    median_rent = "B25064_001"
```

```r
  ),
  year = 2022,
  survey = "acs5"
)
```

```
## Getting data from the 2018-2022 5-year ACS
```

```r
# Clean the Census data so it can join with Zillow data
# Census names include labels like "CDP, Maryland" or "city, Maryland"
# This removes those extra words so the city names match Zillow's RegionName column
census_clean <- census_data %>%
  select(NAME, variable, estimate) %>%
  pivot_wider(names_from = variable, values_from = estimate) %>%
  mutate(
    RegionName = NAME,
    RegionName = str_remove(RegionName, " city, Maryland"),
    RegionName = str_remove(RegionName, " town, Maryland"),
    RegionName = str_remove(RegionName, " village, Maryland"),
    RegionName = str_remove(RegionName, " CDP, Maryland")
  ) %>%
  select(RegionName, population, median_income, census_home_value, median_rent)
head(census_data)
```

```
## # A tibble: 6 x 5
##   GEOID   NAME                                  variable          estimate   moe
##   <chr>   <chr>                                 <chr>                <dbl> <dbl>
## 1 2400125 Aberdeen city, Maryland               population           16422    49
## 2 2400125 Aberdeen city, Maryland               median_income        74555  9128
## 3 2400125 Aberdeen city, Maryland               median_rent           1229   104
## 4 2400125 Aberdeen city, Maryland               census_home_value   227000 13245
## 5 2400175 Aberdeen Proving Ground CDP, Maryland population            2703   540
## 6 2400175 Aberdeen Proving Ground CDP, Maryland median_income       118750 15041
```

```r
# Combine Zillow investment metrics with Census demographic data
# This creates one final dataset with growth, volatility, income, population, rent, and home value
final_summary <- investment_summary %>%
  left_join(census_clean, by = "RegionName") %>%
  mutate(
    # Affordability ratio compares Zillow home values to local median income
    # A higher number means homes are less affordable relative to income
    affordability_ratio = end_price / median_income,

    # Rent-to-income ratio estimates how much of yearly income goes toward rent
    # Median rent is monthly, so it is multiplied by 12
    rent_to_income_ratio = median_rent * 12 / median_income
  )
```

```r
# Create an investment score
# This rewards cities with stronger risk-adjusted growth and better affordability
# Higher score = potentially stronger investment opportunity
final_summary <- final_summary %>%
  mutate(
```

```r
    investment_score = risk_adjusted_return / affordability_ratio
  ) %>%
  arrange(desc(investment_score))

# Display the final summary table
final_summary
```

```
## # A tibble: 32 x 17
##    RegionName   start_date end_date   start_price end_price growth_dollars
##    <chr>        <date>     <date>           <dbl>     <dbl>          <dbl>
##  1 Poolesville  2000-01-31 2026-03-31     262194.   736648.        474454.
##  2 Olney        2000-01-31 2026-03-31     231814.   670386.        438572.
##  3 Garrett Park 2000-01-31 2026-03-31     326681.  1061769.        735088.
##  4 Chevy Chase  2000-01-31 2026-03-31     389743.  1247761.        858019.
##  5 Damascus     2000-01-31 2026-03-31     197062.   560485.        363423.
##  6 Burtonsville 2000-01-31 2026-03-31     187786.   480785.        292999.
##  7 Germantown   2000-01-31 2026-03-31     150852.   423002.        272149.
##  8 Clarksburg   2000-01-31 2026-03-31     243155.   636189.        393034.
##  9 Chevy Chase  2000-01-31 2026-03-31     389743.  1247761.        858019.
## 10 Rockville    2000-01-31 2026-03-31     210841.   618892.        408051.
## # i 22 more rows
## # i 11 more variables: growth_percent <dbl>, avg_monthly_change <dbl>,
## #   volatility <dbl>, risk_adjusted_return <dbl>, population <dbl>,
## #   median_income <dbl>, census_home_value <dbl>, median_rent <dbl>,
## #   affordability_ratio <dbl>, rent_to_income_ratio <dbl>,
## #   investment_score <dbl>
```

```r
# Create a clean final table with the most important variables
# This table helps identify which cities look strongest for investment
final_summary %>%
  select(
    RegionName,
    growth_percent,
    volatility,
    risk_adjusted_return,
    median_income,
    population,
    affordability_ratio,
    investment_score
  ) %>%
  arrange(desc(investment_score)) %>%
  print(n = 20)
```

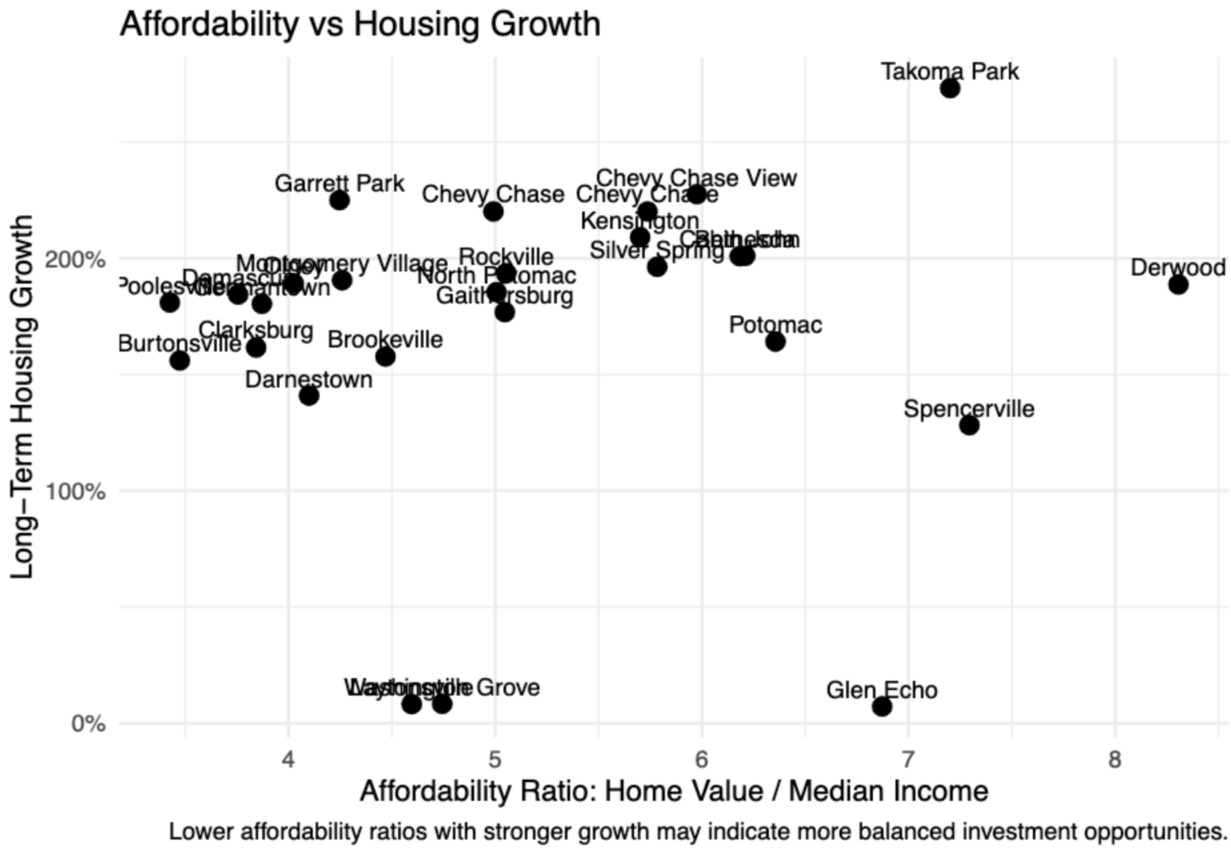
```
## # A tibble: 32 x 8
##    RegionName       growth_percent volatility risk_adjusted_return median_income
##    <chr>                     <dbl>      <dbl>                <dbl>         <dbl>
##  1 Poolesville                1.81    0.00702                 258.        215025
##  2 Olney                      1.89    0.00697                 271.        166541
##  3 Garrett Park               2.25    0.00795                 283.        250001
##  4 Chevy Chase                2.20    0.00694                 317.        250001
##  5 Damascus                   1.84    0.00780                 236.        149234
##  6 Burtonsville               1.56    0.00758                 206.        138416
##  7 Germantown                 1.80    0.00793                 227.        109268
```

```
##  8 Clarksburg                   1.62    0.00758                    213.          165551
##  9 Chevy Chase                  2.20    0.00694                    317.          217500
## 10 Rockville                    1.94    0.00699                    277.          122470
## 11 North Potomac                1.85    0.00680                    273.          182083
## 12 Chevy Chase View             2.28    0.00726                    313.          250001
## 13 Kensington                   2.09    0.00706                    296.          143021
## 14 Montgomery Vill~             1.91    0.00868                    220.           91703
## 15 Takoma Park                  2.73    0.00749                    365.           95316
## 16 Brookeville                  1.58    0.00712                    222.          182188
## 17 Darnestown                   1.41    0.00729                    193.          250001
## 18 Bethesda                     2.01    0.00688                    292.          185546
## 19 Gaithersburg                 1.77    0.00747                    237.          104544
## 20 Cabin John                   2.01    0.00732                    274.          208598
## # i 12 more rows
## # i 3 more variables: population <dbl>, affordability_ratio <dbl>,
## #   investment_score <dbl>
```

```r
# Visualize affordability compared to housing growth
# This helps show whether high-growth cities are still affordable relative to income
ggplot(final_summary, aes(x = affordability_ratio, y = growth_percent, label = RegionName)) +
  geom_point(size = 3) +
  geom_text(vjust = -0.5, size = 3) +
  scale_y_continuous(labels = percent_format()) +
  labs(
    title = "Affordability vs Housing Growth",
    x = "Affordability Ratio: Home Value / Median Income",
    y = "Long-Term Housing Growth",
    caption = "Lower affordability ratios with stronger growth may indicate more balanced investment op
  ) +
  theme_minimal()
```

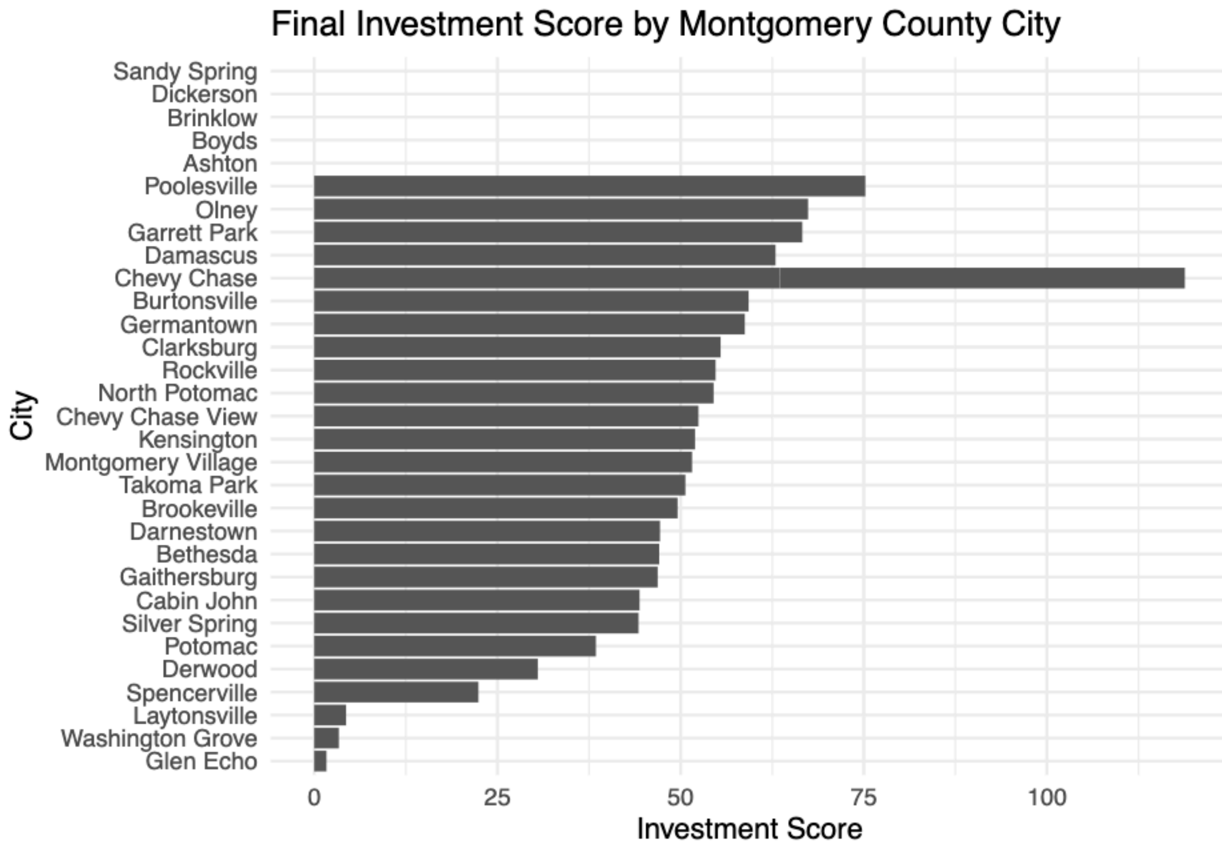
```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_text()`).
```




```
# Visualize the final investment score by city
# This chart ranks cities using growth, volatility, and affordability together
ggplot(final_summary, aes(x = reorder(RegionName, investment_score), y = investment_score)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Final Investment Score by Montgomery County City",
    x = "City",
    y = "Investment Score"
  ) +
  theme_minimal()
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_col()`).
```




```r
# Remove cities that did not successfully match with Census data
# This prevents missing income or population values from affecting the final ranking
final_summary_clean <- final_summary %>%
  filter(
    !is.na(median_income),
    !is.na(population),
    !is.na(affordability_ratio),
    !is.na(investment_score)
  )

final_summary_clean
```

```
## # A tibble: 27 x 17
##    RegionName   start_date end_date   start_price end_price growth_dollars
##    <chr>        <date>     <date>           <dbl>     <dbl>          <dbl>
##  1 Poolesville  2000-01-31 2026-03-31     262194.   736648.        474454.
##  2 Olney        2000-01-31 2026-03-31     231814.   670386.        438572.
##  3 Garrett Park 2000-01-31 2026-03-31     326681.  1061769.        735088.
##  4 Chevy Chase  2000-01-31 2026-03-31     389743.  1247761.        858019.
##  5 Damascus     2000-01-31 2026-03-31     197062.   560485.        363423.
##  6 Burtonsville 2000-01-31 2026-03-31     187786.   480785.        292999.
##  7 Germantown   2000-01-31 2026-03-31     150852.   423002.        272149.
##  8 Clarksburg   2000-01-31 2026-03-31     243155.   636189.        393034.
##  9 Chevy Chase  2000-01-31 2026-03-31     389743.  1247761.        858019.
## 10 Rockville    2000-01-31 2026-03-31     210841.   618892.        408051.
## # i 17 more rows
```

```
## # i 11 more variables: growth_percent <dbl>, avg_monthly_change <dbl>,
## #   volatility <dbl>, risk_adjusted_return <dbl>, population <dbl>,
## #   median_income <dbl>, census_home_value <dbl>, median_rent <dbl>,
## #   affordability_ratio <dbl>, rent_to_income_ratio <dbl>,
## #   investment_score <dbl>
```

```r
# Final ranked table of strongest investment areas
# This table is the main output of the project
top_investment_cities <- final_summary_clean %>%
  select(
    RegionName,
    growth_percent,
    volatility,
    risk_adjusted_return,
    median_income,
    population,
    affordability_ratio,
    investment_score
  ) %>%
  arrange(desc(investment_score))

top_investment_cities
```

```
## # A tibble: 27 x 8
##    RegionName   growth_percent volatility risk_adjusted_return median_income
##    <chr>                 <dbl>      <dbl>                <dbl>         <dbl>
##  1 Poolesville            1.81    0.00702                 258.        215025
##  2 Olney                  1.89    0.00697                 271.        166541
##  3 Garrett Park           2.25    0.00795                 283.        250001
##  4 Chevy Chase            2.20    0.00694                 317.        250001
##  5 Damascus               1.84    0.00780                 236.        149234
##  6 Burtonsville           1.56    0.00758                 206.        138416
##  7 Germantown             1.80    0.00793                 227.        109268
##  8 Clarksburg             1.62    0.00758                 213.        165551
##  9 Chevy Chase            2.20    0.00694                 317.        217500
## 10 Rockville              1.94    0.00699                 277.        122470
## # i 17 more rows
## # i 3 more variables: population <dbl>, affordability_ratio <dbl>,
## #   investment_score <dbl>
```

Conclusion

This project analyzed Montgomery County housing markets using Zillow ZHVI data and Census ACS demographic data. The analysis compared long-term housing growth, volatility, median income, population, and affordability.

The results show that the strongest potential investment areas are not always the most expensive cities. Instead, stronger areas are those that combine long-term price growth with lower volatility and stronger local income support.

The final investment score helps rank each city by combining risk-adjusted housing growth with affordability. This provides a more balanced view of investment potential than looking at home prices alone.

#Limitations

This analysis uses Zillow ZHVI estimates rather than actual transaction-level sales data. Census ACS data is also based on survey estimates, and some Census place names may not perfectly match Zillow city names. The project does not include property taxes, interest rates, crime, school quality, rental vacancy, or cap rates, which would improve a full real estate investment analysis.